

COMP3018: Set Exercises; Human-Robot Interaction (HRI)

Cultural Differences and Probabilistic Modelling in Human-Robot Interaction

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1- Task (1): Cultural Differences and HRI Design

1.1. Cultural Differences in the Acceptance of Robots (Kaplan, 2004)

Kaplan (2004) identifies a two-way East-West societal divergence: “Culture affects the way technology is perceived and, in a reciprocal manner, technological evolution shapes culture in particular ways.” (.pdf-p. 1); i.e. the cultural, theological, and mythological narrative of each region shapes the societal meta-layer (*the deeper cultural wiring controlling how a society receives robots*).

1.1.1 Western Society (The Frankenstein Syndrome)

Western culture has persistently viewed the creation of human-like (humanoid) entities quite suspiciously. Kaplan (2004) identifies this as the “Frankenstein syndrome”: any artificially created humanoid will necessarily turn against his creator at some point’ (p. 11, section-4.3, line-7).

This anxiety traces to the West’s distinction between nature and culture, which posits “no place for hybrids” in such classifications (Kaplan, 2004, p. 6, section-3.2 line-21). The Western cultural narrative therefore frames humanoid robots as a challenge to human specificity (p. 14, section-5.2, lines-4-6), and thus, a transgression (*violation of the boundary between what humans create, and what humans are*) against the natural order; as a result, the convergence of humans and machines remains “both fascinating and frightening” (Kaplan, 2004, p. 15, section-6, lines-36-37), wherein the machine is culturally positioned as a functional instrument rather than a would-be social entity. Kaplan further notes the concept of “narcissistic shields” (*the psychological-defence mechanisms protecting human exceptionalism*) (p. 14, section-5.2, lines-10-11), whereby Westerners psychologically distance themselves from machines that erode the human-robot distinction.

1.1.2 Eastern Society (Technology Taming and Animism)

Japanese culture exhibits a fundamentally different ontological (*the definition of what counts as a being*) stance. Kaplan (2004) traces this to the Shinto tradition (p. 5, section-3, lines-2-11), wherein the rigid Western boundary between animate and inanimate is dissolved in favour of what Kaplan describes as a “continuous network of beings” (p. 6, section-3.2, lines-24-25). In this view, humanoid robots are not perceived as transgressions but instead as natural extensions.

Furthermore, Kaplan discussed the cultural mechanism of “technology taming” i.e. a recurring historical pattern wherein foreign technologies are domesticated via integration into existing cultural frameworks (p. 2, section-1, lines-8-9). This ethos aligns with the *kata* tradition of formalised practice, where repetition leads to “maximum stability” (p. 6, section-3.1, line-4). The popular *Astro Boy* manga franchise (p. 2, section-2.1, lines-24-35) exemplifies this domestication narratively. Kaplan references the Amaterasu myth (p. 5, section-3, lines-2-11) to argue that Japanese cosmology fundamentally lacks the creator-vs-creation antagonism (the inherent transgression of usurping divine privilege) that underpins Western technophobia, saying simply that “in Japan, no gods created human beings” (p. 12, section-4.4, line-20). This cultural openness persists into the contemporary era; as established in the module materials, Japanese society is “probably less sensitive to risks that might appear from robots” than the West (Lecture 5).

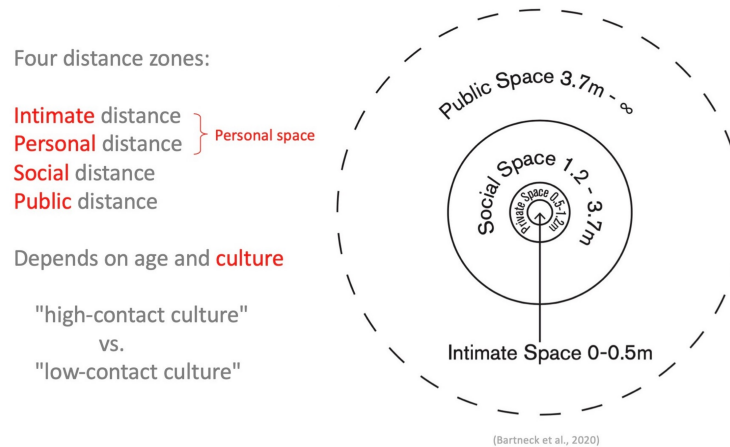
1.1.3 Implications for HRI Design

The divergent cultural framings dictate interaction design profoundly within HRI; whilst Western persons predominantly prefer robots that maintain clear machine identity markers, thereby preserving the ‘narcissistic shield’ (Kaplan, 2004, p. 14, section-5.2, lines-10-11), Eastern users instead welcome human-like (anthropomorphic) features that align with the animistic expectations established in Section 1.1.2; Lim, Rooksby and Cross (2021, p. 1321) observed this contrasting preference, confirming that culture significantly influences the acceptance of robotic morphology as Korean participants envisioned human-like robots serving as “social company,” whereas US participants instead envisioned theirs as “machine-like” extensions of “household appliances”.

Whilst this is true, a Western robot-designer may be inclined to impose universal proxemic standards (culturally-defined, personal-space boundaries) based on their own norms. However, non-contact cultures

such as Japan maintain larger personal-space buffers, whereas contact cultures such as those in Southern Europe tolerate closer approach distances (Joosse, Lohse and Evers, 2014, pp. 1-2); therefore, a culturally-calibrated model should feed local boundaries into its approach-vector calculations, as failing to respect these bounds contravenes user expectations, effectively alienating the would-be companion (Rios-Martinez, Spalanzani and Laugier, 2015, p. 140).

Proxemics



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Figure 1: Lecture-4 slide illustrating Hall's four proxemic distance zones and their cultural dependence.

Designers should also account for the novelty effect (Figure-2); users for whom the robot represents an entirely novel stimulus exhibit inflated acceptance ratings. The novelty effect, conventionally framed as “a source of noise in need of reduction” and “behavioural disturbances” obscuring the phenomenon under investigation (Smedegaard, 2019, pp. 411-412; Lecture 6), thus confounds true cultural preference with transient unfamiliarity. The operative principle is therefore cultural relativism: researchers cannot infer permanent acceptance from early data; instead, the system needs-to evaluate engagement over time.

2.2.2 Novelty Effect

- Novelty effects can be particularly strong in HRI
 - Many participants may have never seen or interacted with a robot before.
 - We don't want the effects of this first interaction (so-called novelty effects) to completely overpower the manipulation of the independent variable in our study.
- One way to reduce impact:
 - Include a practice session with the robot as part of your study
 - Data from this practice session would not be recorded or used in our analysis.

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Figure 2: Lecture-6 slide defining the novelty effect in HRI and its mitigation via practice sessions.

1.2. African Cultural Factors Influencing HRI Design

Kaplan's (2004, Abstract) *confines* the analysis to the East-West axis, framed explicitly as an inquiry into whether robots are "perceived in the same manner in the West and in Japan"; however, a complete account should address the African context.

1.2.1 Ubuntu Philosophy and Communal Identity:

Ubuntu: a Southern African philosophical principle that "a person is a person through other persons" (Metz, 2007, p. 323). Whereas Western HRI design foregrounds individual user experience (Lim, Rooksby and Cross, 2021, p. 1325), Ubuntu-driven design would prioritise communal benefit and relational harmony. A robot operating within an Ubuntu-oriented society should therefore be designed to address the *group* rather than the individual, facilitating collective decision-making and shared resource access; an inference drawn from Metz's (2007, pp. 324-326) emphasis on consensus over dissent and communal resource distribution. This contrasts with the individualised personal-assistant paradigm prevalent in Western HRI.

1.2.2 Power Distance and Hierarchical Norms:

Cirasa and Conti's (2025, p. 7) scoping review identifies Hofstede's cultural dimensions framework (a model measuring societal values across indices such as Power Distance and individualism) as dimensions that "significantly affect interactions with technology". Addressing the literature gap outside Western and East Asian spheres, Power Distance can be applied to the African context where societies such as Nigeria and Ghana score high on this dimension, such that hierarchical authority structures strictly govern social interaction.

In regards to HRI, this suggests that robots interacting with users across different social strata should modulate behaviours accordingly (Cirasa and Conti, 2025, p. 7): deferential language and posture when addressing elders or authoritative figures; a more directive interaction-style when assisting in contexts where the robot is perceived as an institutional representative. Failing to encode these hierarchical norms risks violating deeply held social expectations, thereby undermining trust (Hancock et al., 2011, p. 518).

1.2.3 Oral Tradition and Multimodal Communication:

African cultures historically have preferred oral knowledge transmission over written documentation (Winschiers-Theophilus and Bidwell, 2013, *.pdf's*-pp. 12-13). This implicates interaction modality, i.e. voice-driven, narrative-based interfaces using speech processing and *prosodic* (tonal) features such as pitch and MFCCs (Lecture 3), may achieve higher engagement than text-heavy GUI paradigms. Furthermore, Lecture 2 established that 65% of daily-life communication is nonverbal; thus, gestural and paralinguistic channels become critical design considerations for African contexts wherein oral expressiveness is culturally normative.

1.2.4 Infrastructure and Access Constraints:

Despite rapid technological growth, many African regions face infrastructure limitations, e.g. intermittent connectivity and limited access to high-specification hardware (Wyche and Steinfield, 2016, p. 327). HRI systems deployed in these contexts should therefore be robust to connectivity loss, operable on low-power devices, and designed for shared rather than personal ownership, aligning with the communal ethos of Ubuntu.

1.3 Regional Design Traits (Appearance and Behaviour)

The cultural factors identified above dictate distinct morphological-and-behavioural traits (how the robot looks and how it acts) to maximise acceptance (Fong, Nourbakhsh and Dautenhahn, 2003, p. 149).

(a) The East (Japan). Because Shinto animism dissolves the natural/artificial boundary (Kaplan, 2004, p. 6, section-3.2, lines-14-15), anthropomorphic or highly expressive aesthetic traits are welcomed. However, to align with the *kata* tradition of harmonious form (Kaplan, 2004, p. 6, section-3.1, lines-2-4), the robot's movements need to be fluid and graceful rather than purely functional. Behaviourally, the robot should adopt a "side-by-side" cooperative posture rather than an imposing face-to-face stance, reflecting Japanese non-tactile proxemic norms requiring larger personal-space buffers (Joosse, Lohse and Evers, 2014, p. 2 & Lecture 4). As Lecture 2 establishes, *haptics* (deliberate physical communication) is replaced by proxemic attentiveness in non-tactile cultures, maintaining Hall's personal zone of 0.45-1.2m (Rios-Martinez, Spalanzani and Laugier, 2015, p. 140).

(b) The West (Europe/North America). To avoid triggering the Frankenstein Syndrome and to respect the "narcissistic shield" (Kaplan, 2004, p. 14, section-5.2, lines-10-11), Western robots should possess functional, machine-like aesthetic markers (visible joints, metallic surfaces) rather than overtly human-like features, thereby avoiding descent into the uncanny valley (Mori, 1970, p. 98). Behaviourally, they should exhibit transparency, explicitly stating operational reasoning to alleviate what Kaplan (2004, p. 11, section-4.3, lines-16-17) characterises as anxieties of autonomous transgression. The handshake serves as a *symbolic gesture* (a movement with a culturally agreed-upon meaning) combined with *haptics* (Lecture 2), albeit calibrated to the Mediterranean vs Northern European proxemic distinction (Lecture 4).

(c) Africa. Informed by Ubuntu and high Power Distance (Cirasa and Conti, 2025, p. 7), an African-deployed robot should possess a modest physical stature to avoid perceived challenges to human hierarchical authority. Behaviourally, it should be group-facing rather than dyadic (one-robot-to-one-user), utilising a warm, highly expressive vocal synthesiser capable of rendering the rich prosodic variations (pitch, tone) necessary for an oral-tradition society (Lecture 3; Winschiers-Theophilus and Bidwell, 2013, *.pdf's*-pp. 12-13). Furthermore, Lecture 2 established that "65% of communication is non-verbal," gestural and paralinguistic channels, specifically *beat gestures* (rhythmic hand movements accentuating speech rhythm) and *iconic gestures* (movements visually representing the subject), become critical design considerations.

1.4. Adapting Design Patterns for Sociality (Kahn et al., 2008)

Kahn et al. (2008) identify eight design patterns for sociality in HRI, noting the patterns are "likely under-described" (Section 4). These patterns should be regionally adapted using the traits established in Section 1.3:

1.4.1 Pattern 1: Initial Introduction (Kahn et al., 2008, p. 100)

- (a) *East*: Rather than tactile handshakes, the robot should initiate interaction with a calibrated bow, as “in Japan it’s very considered impolite if you break the personal distance or space and try to touch somebody” (Lecture 4). The bow angle should parametrically encode social hierarchy recognition.
- (b) *West*: The *Initial Introduction* can incorporate the handshake as a tactile greeting, albeit with sensitivity to the Mediterranean (closer) versus Northern European (distant) proxemic distinctions (Lecture 4).
- (c) *Africa*: Reflecting Ubuntu’s communal orientation, the introduction should address groups rather than individuals. Encoding Power Distance norms, the robot should always greet the eldest or most-senior member first; an inference drawn from the high Power Distance norms identified by Cirasa and Conti (2025, p. 7).

1.4.2 Pattern 4: Personal Interests and History (Kahn et al., 2008, p. 101)

- (a) *East*: The robot’s backstory can elaborately integrate into the animistic expectation of objects possessing a “spirit” or character (Kaplan, 2004, p. 6, section-3.2, lines-22-25).
- (b) *West*: Self-disclosure needs-to be transparently mechanical, framing its “interests” around its programmed purpose to reinforce the user’s ontological comfort (Lecture 1; the Western-“Utilitarian”-robot-as-tool stance).
- (c) *Africa*: This pattern should be realised via storytelling and voice-based dialogue. The robot should establish *joint attention* (the ability of multiple agents to focus on a shared reference point) via structured eye gaze to maintain narrative authority (Lecture 2).

1.4.3 Pattern 5: Recovering From Mistakes (Kahn et al., 2008, p. 101)

- (a) *East*: The robot should employ indirect acknowledgement strategies that preserve social harmony (Kaplan, 2004, p. 6, section-3.2, lines-22-25), avoiding direct self-criticism that may cause discomfort via loss of face.
- (b) *West*: The robot should explicitly explain its error and how it will correct it, prioritising transparency to mitigate the underlying technophobia Kaplan (2004, p. 11, section-4.3, lines-5-8) identifies.
- (c) *Africa*: Error recovery needs-to be calibrated to social rank; an unacknowledged error affecting an elder constitutes a severe social violation, and thus recovery should involve deferential posture and movement (Lecture 2) to protect the user’s social face within the community.

1.5 Summary of Cultural Design Implications

Table 1: Cultural factors and corresponding HRI design adaptations.

Region	Key Cultural Factor	Design Adaptation
East (Japan)	Shinto animism; non-tactile norms; <i>kata</i> tradition	Bowing protocols; enlarged personal-space buffers; indirect error recovery
West	Frankenstein Syndrome; individualism; narcissistic shields	Machine-identity markers; <i>handshake-ready</i> ; transparent self-disclosure
Africa	Ubuntu; high Power Distance; oral tradition	Communal greetings; elder-first hierarchy; <i>voice-driven narrative interfaces</i>

Kahn et al. (2008) conclude: effective HRI must be “compelling as a lived experience” (p. 103) and thus what constitutes a compelling experience is, as demonstrated above, inseparable from the cultural context thereof.

2- Task (2): POMDPs in Human-Robot Interaction

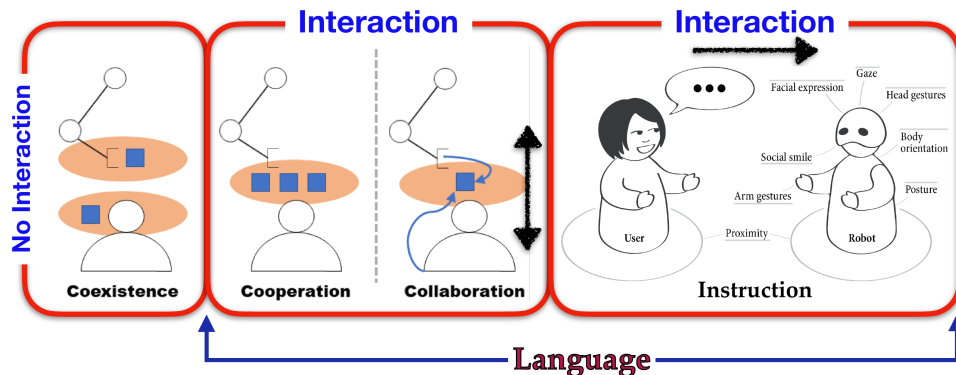
2.1 The Role of POMDPs in Trust, Cooperation, Coordination, and Collaboration

A Partially Observable Markov Decision Process (POMDP) extends the MDP introduced in *COMP3003* (Lecture 7) by relaxing full-state observability: the robot can only *infer* the true state via noisy, incomplete observations (Kaelbling, Littman and Cassandra, 1998, Section-3.1, p. 105). Formally, a POMDP is defined by the tuple $\langle S, A, T, R, \Omega, O, \gamma \rangle$, wherein S is a finite set of states, A the available actions, $T(s, a, s') = P(s' | s, a)$ the transition function, $R : S \times A \rightarrow \mathbb{R}$ the reward function, Ω a finite set of observations, $O(s', a, o) = P(o | s', a)$ the observation function and $\gamma \in [0, 1)$ the discount factor.

To understand the POMDP's utility, one needs to *differentiate interaction paradigms* as defined in Lecture 1 (Figure~3). **Coexistence** is where agents share an environment but complete different tasks, requiring only fully-observable physical states to avoid collisions. **Cooperation** involves a shared workspace and complementary parts of a task. **Coordination** requires temporal alignment: sequencing complementary actions so neither agent blocks nor duplicates the other; the POMDP's belief state enables this by allowing the robot to anticipate *when* to act versus yield based on inferred human intent. However, true **collaboration** demands a shared workspace and the *exact same shared goal* (Lecture 1). In collaboration, the robot has to continuously align its actions with the human's unobservable mental states: trust, intent, cognitive load. Because the robot is fundamentally blind to these latent variables, the POMDP's belief state b becomes the computational prerequisite for graduating from mere coexistence to true collaboration (Chen et al., 2020, p. 6). Nikolaidis et al. (2017, pp. 621-623) indeed demonstrate this empirically via a “Bounded-Memory Adaptation Model” (BAM) wherein the robot maintains a mixed-observability MDP over the human's latent adaptability, showing mutual adaptation via belief-space planning hugely outperforms fixed strategies.

Human-Robot Interaction

- **Human-Robot Interaction** can be characterized in terms of interdependency by:
 - **Coexistence**: same environment & different tasks (no “or minimal” interaction)
 - **Cooperation**: same workspace & complementary parts of a task (shared goal)
 - **Collaboration**: same workspace & same task (shared goal)
 - **Instruction**: same environment (workspace) & master-slave setting



- [1] E. Matheson, R. Minto, E. G. G. Zampieri, M. Faccio and G. Rosati, “Human-Robot Collaboration in Manufacturing Applications: A Review”, *Robotics*, 8(4), 2019
- [2] J. Butepage and D. Kragic, “Human-Robot Collaboration: From Psychology to Social Robotics”, arXiv:1705.10146, 2017
- [3] B. Mutlu, “Designing Embodied Cues for Dialog with Robots”, *AI Magazine*, 32(4):17–30, 2011

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Figure 3: Lecture-1 slide defining the four HRI interaction paradigms: coexistence, cooperation, collaboration, and instruction. The POMDP framework is necessary for graduating beyond coexistence, as collaboration requires modelling the human's unobservable mental states.

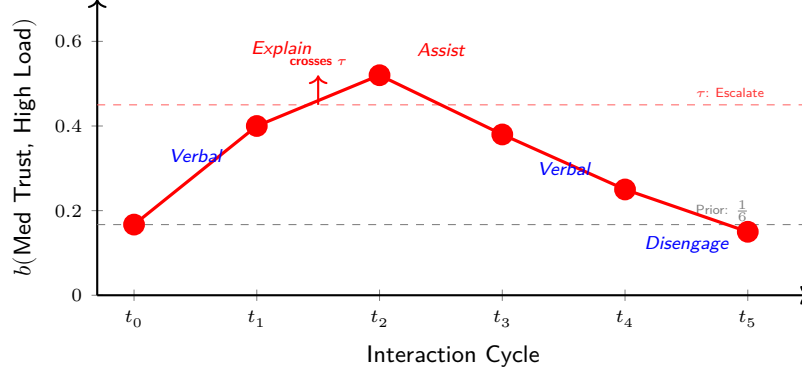


Figure 4: Action selection governed by continuous belief thresholds. When the probabilistic belief $b(\text{Med Trust, High Load})$ exceeds the escalation threshold $\tau = 0.45$ at t_2 , the policy dictates a discrete shift from Verbal_Remind to Explain_Benefits, then Offer_Physical_Assist. As trust recovers and the belief drops below τ (from t_3), the robot de-escalates accordingly.

2.2 Uncertainty, Belief States, and Decision-Making

Because the true state is hidden, the POMDP agent maintains a **belief state** b : a probability distribution over all possible states S , where $b(s)$ represents the agent’s subjective probability that the environment is in state s , such that $\sum_{s \in S} b(s) = 1$. After taking action a and receiving observation o , the belief state is updated via **Bayesian filtering**:

$$b'(s') = \eta \cdot O(s', a, o) \sum_{s \in S} T(s, a, s') \cdot b(s)$$

wherein η is a normalisation constant ensuring $\sum_{s'} b'(s') = 1$. This update rule captures the epistemic challenge of HRI: the robot must continuously revise its model of the human’s internal state as new and could-be contradictory evidence arrives. As Lecture 3 establishes, the robot utilises affective-computing descriptors (e.g. pitch, MFCCs, zero-crossing rate) to extract observations from the human’s behaviour, *feeding* these into the belief update.

The belief state b therefore serves as a **sufficient statistic** for the entire interaction history, compressing all past actions and observations into a single probability vector (Kaelbling, Littman and Cassandra, 1998, Section-3.2, p. 106). This is why the POMDP exhibits a non-Markovian *policy* (each action depends on the full history h_t) whilst maintaining a Markovian *belief update* (the next belief b' depends only on the current belief b , the action taken, and the observation received); the belief state absorbs all relevant history, and thus the agent need not store raw trajectories to act optimally.

2.3. Challenges of Trust Modelling and the POMDP Response

Trust is a latent psychological variable; it cannot be directly measured, only inferred from observable behavioural indicators. Lee and See (2004, *.pdf*-p. 6) define trust as “the attitude that an agent will help achieve an individual’s goals in a situation characterized by uncertainty and vulnerability”; a definition foregrounding the latent nature that necessitates probabilistic modelling. Three challenges arise.

2.3.1 The Measurement Problem

Observations such as compliance rate, response latency, gaze direction (Lecture 2), and verbal affirmations are noisy proxies (unreliable indirect measures). Hancock et al.’s (2011, p. 520) meta-analysis of 29 empirical studies confirms this, finding that robot performance-based factors exhibit the strongest correlation with trust (mean $r = +0.34$, p. 522), yet even these explain only modest variance; a user may comply with a

robot’s suggestion despite low trust (e.g. time pressure), or indeed refuse despite high trust (e.g. due to task complexity), and thus the observation alone does not necessarily disambiguate the latent state.

2.3.2 Temporal Dynamics

Trust evolves non-linearly, as it builds slowly through consistent performance but degrades rapidly after errors. This asymmetry is empirically confirmed by Desai et al. (2013, p. 256), who found that “the recovery of trust after a reliability drop occurs at a slower pace than the pace at which trust develops before reliability drops”. The *Recovering From Mistakes* pattern (Kahn et al., 2008, p. 101) is therefore critical: a robot that acknowledges and corrects errors can arrest trust decay, whereas one that ignores failures risks appearing aggressive.

2.3.3 Computational Intractability

Solving POMDPs exactly is PSPACE-complete (Papadimitriou and Tsitsiklis, 1987, p. 448), as the belief simplex is continuous even with finite $|S|$. Practical HRI applications therefore utilise approximate solvers such as PBVI (Pineau, Gordon and Thrun, 2003, p. 1025) or POMCP (Silver and Veness, 2010, p. 4) for tractable real-time planning.

The POMDP addresses these challenges by encoding trust as a hidden state variable, observations as probabilistic signals thereof, and actions as trust-modulating strategies (Chen et al., 2020, p. 6); trust is therefore modelled as a continuously-evolving distribution rather than a threshold-based heuristic (a rigid binary switch ignoring subtle fluctuations).

2.4. Proposed Neuro-Symbolic POMDP Model: Neo Robot with OpenAI Cognitive Architecture

To concretise this framework, I propose a neuro-symbolic POMDP model for a Neo (Pepper) humanoid robot augmented with an OpenAI multimodal API, deployed as an elderly medication-adherence assistant. LLMs are powerful observation extractors, parsing unstructured human behaviour into structured probabilistic assessments; however, they are inherently stateless (each API call is independent, with no temporal memory) and prone to hallucination (Ji et al., 2023, p. 36). Wrapping the API inside a POMDP belief state $b(s)$ therefore provides the temporal scaffold the LLM lacks: a continuously-updated model of the human’s latent states (Trust, Cognitive Load). This neuro-symbolic paradigm, wherein a neural subsystem handles perception whilst a symbolic subsystem governs reasoning, represents what Garcez and Lamb (2023, p. 12389) term the ‘third wave’ of AI. Ahn et al. (2022, p. 4) established via SayCan that LLMs can ground language in robotic affordances, yet their architecture lacks the temporal belief maintenance a POMDP provides.

2.4.1 The Neuro-Symbolic Architecture

The Neo robot’s onboard sensors (camera, microphone array, tactile sensors) capture multimodal data from the elderly user, transmitted to the OpenAI API serving as the **observation function** (O). The API processes this stream, analysing facial action units, extracting prosodic descriptors (pitch, MFCCs, zero-crossing rate) from speech (Lecture 3), and interpreting gestural semantics, to output a structured observation $o \in \Omega$. The API can be prompted to estimate a probability distribution over possible observations rather than a categorical label, preserving the epistemic uncertainty the POMDP requires. If the robot performs *Verbal_Remind* and the API observes *Hesitate*, the belief shifts toward medium-trust, high-load states; $\pi^*(b)$ consequently selects *Explain_Benefits* rather than escalating to physical assistance (Figure~4), as the belief indicates the user is cognitively loaded rather than non-compliant.

2.4.2 Formal Specification

- **State Space** (S): Trust $\in \{\text{Low, Medium, High}\} \times \text{Cognitive Load} \in \{\text{Low, High}\}$, yielding $|S| = 6$.
- **Action Space** (A): $\{\text{Verbal_Remind, Explain_Benefits, Offer_Physical_Assist, Increase_Autonomy, Disengage}\}$. The POMDP selects abstract actions; the API translates these into culturally-calibrated language and gestures via the Neo robot.

- **Observation Space (Ω):** {Comply, Hesitate, Verbal_Refuse, Ignore, Gaze_Avert}, extracted by the OpenAI API.
- **Observation Function (O):** $P(o \mid s', a)$ is estimated by the API’s multimodal inference; e.g. $P(\text{Comply} \mid \text{HighTrust}, \text{LowLoad}, \text{Remind}) = 0.8$, derived from the user’s facial configuration and vocal tone.
- **Transition Function (T):** Models trust dynamics parametrically. An unneeded Offer_Physical_Assist violating personal proxemics (Lecture 4) degrades trust: $P(\text{Low} \mid \text{Med}, \text{Assist}) = 0.6$; conversely, a well-timed Explain_Benefits yields $P(\text{High} \mid \text{Med}, \text{Explain}) = 0.5$.
- **Reward Function (R):** Successful medication adherence yields $R = +10$; preserving user autonomy (choosing Increase_Autonomy when trust is High) yields $R = +3$; unwanted physical assistance incurs $R = -5$, reflecting the social cost of proxemic violation. Mirroring the POMDP’s ‘cost of listening’ penalty (the Tiger problem), the architecture imposes $R = -1$ per observation cycle, penalising excessive polling. Both O and R should be culturally parametrised from the findings in Task 1; the proxemic penalty, for instance, should be weighted more heavily for non-contact cultures.

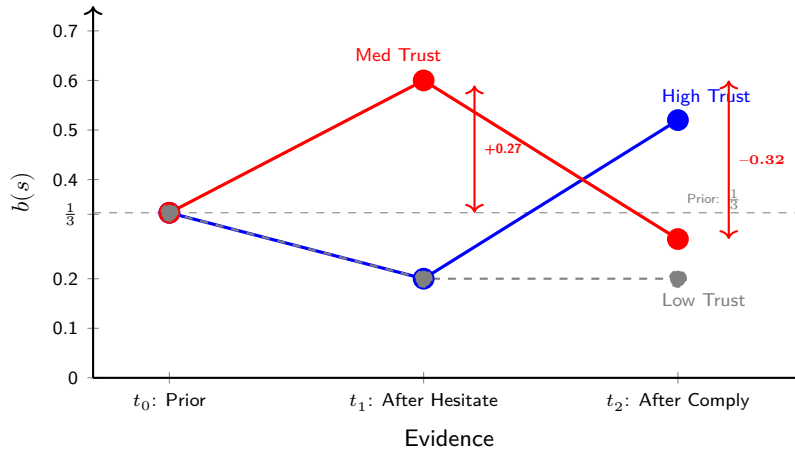


Figure 5: Simplified belief evolution over a three-state trust space. The medium-trust posterior exhibits non-monotonic behaviour, rising sharply after observing Hesitate at t_1 , then declining upon Comply at t_2 . These deviations from the uniform prior demonstrate the Bayesian update mechanism inferring hidden trust states from partial observations, consistent with the non-linear trust dynamics described by Lee and See (2004, .pdf-p. 22).

2.4.3 Benefits and Limitations

The model’s strength lies in neuro-symbolic complementarity (Garcez and Lamb, 2023, p. 12389). Unlike traditional solvers such as PBVI (Pineau, Gordon and Thrun, 2003, p. 1025), this approach delegates perceptual dimensionality-reduction to the LLM, preserving belief-update tractability. However, the architecture introduces limitations: 1) a fundamental mathematical friction: the POMDP relies on the Markov property with a stationary observation function $O(s', a, o)$, yet the LLM is inherently non-stationary; its outputs depend on a dynamic context window, meaning the observation function shifts based on the LLM’s own internal states, and thus the belief update is technically an approximation rather than an exact sufficient statistic. If the LLM’s interpretation silently shifts (e.g. following an API version update), the belief state degrades without any mechanism to detect this drift; 2) API latency (200-800ms) may disrupt real-time proxemic responsiveness (Lecture 4); 3) the LLM’s stochastic nature means identical inputs may yield different observation distributions, introducing unmodelled noise into the Bayesian update; and 4) the observation model may inherit training-data biases, degrading inference accuracy for elderly users if predominantly trained on younger demographics (Lecture 5).

2.5 Ethical and Social Implications

By operationalising trust as a now-computable metric, the POMDP risks enabling exploitation: insofar as the reward function solely prioritises compliance, the optimal policy may learn to time requests when inferred cognitive load is highest. Designers should therefore encode user autonomy into the reward structure, lest assistance erode into manipulation. Sharkey (2014, pp. 69-70) frames this via the Capability Approach: the robot must expand rather than impede access to Nussbaum's central capabilities, and thus the $R = +3$ autonomy bonus is not merely a design preference but an ethical imperative encoding that dignity requires preservation of choice.

The OpenAI API introduces three additional ethical dimensions. Firstly, **hallucination risk**: LLMs generate plausible but factually incorrect outputs (Ji et al., 2023, p. 36); in a medication-adherence context, misclassifying confused hesitation as willing compliance could trigger inappropriate actions with health consequences. Explainable AI (Lecture 5) therefore becomes non-negotiable: the POMDP should justify *why* it selected a particular action, tracing the decision through the belief state to the observations extracted. Secondly, **cloud-data sovereignty**: continuous multimodal processing transmitted to third-party servers risks what Sharkey and Sharkey (2012, pp. 35-36) identify as monitoring that infringes on “the right to privacy” (Figure~6); the user's most vulnerable moments are thereby streamed to servers whose data-handling policies cannot be meaningfully audited. Finally, **API latency and proxemic violation**: response delays during a time-critical approach cause the robot to freeze or fail to yield, behaviour that Lecture 4 identifies as “aggressive” (Figure~7). The ethical watchword is therefore proactive regulation (Lecture 5): designers need to encode transparent, auditable constraints *ex ante* into both the reward function and the API pipeline, ensuring the robot does not merely *model* trust but instead *earns* it through explainable, privacy-preserving behaviour.

Some Issues Raised due to the Use of Technology



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Figure 6: Lecture-5 slide raising the surveillance concern: “The AI records what you do and transfers data... to whom? Company? Third Party?”

Socially Appropriate Positioning

Normally, humans yield to each other when approaching

- To avoid entering each other's personal space
- Using nonverbal signals

Is this culturally dependent?

A robot that doesn't do this may appear aggressive

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Figure 7: Lecture-4 slide establishing that a robot which fails to yield during spatial approach “may appear aggressive”; with the annotation “Is this culturally dependent?”

Appendices

Appendix A: References (Peer-reviewed or Conference only)

Task (1)’s

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Appendix B: AI Declaration

Solo Work	S1 - Generative AI tools have not been used for this assessment.
Assisted Work	A1 – Idea Generation and Problem Exploration Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student's responsibility.
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.
Partnered Work	A11 - Other uses not listed above Please specify:
	P1 - Generative AI tool usage has been used integrally for this assessment Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:

Figure 8: Student Declaration of AI Tool use in this Assessment Table

I declare that I've used the AI tools listed below whilst preparing this assessment. I've read and understood the University of Plymouth's policy on the use of AI tools in assessment and confirm that my use falls within

the coursework's allowed categories, i.e. **A2 (Planning and Structuring Projects)** and **A4 (Research Assistance)**.

AI Tool Used	Purpose of Use	Extent of Use
ChatGPT	Finding relevant pages to read in the paper (A4)	If the paper takes too long to consume efficiently
ChatGPT	General conversations via web-search AI about how the paper's topic relates to others' studies (A4)	Few times
ChatGPT	Used to identify which sections were over-weighted relative to mark allocation for trimming to match word-count allowance (A2)	Once
ChatGPT	Traversing papers to find relevant parts to read for the essay (A4)	Quite a few times

- ☒ I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.
- ☒ I confirm that all details provided above are an accurate description of how AI was used for this assessment.